METABOLIC & FAT LOSS

5-Amino-1MQ — the full review

SMALL-MOLECULE NNMT INHIBITOR

This is the deeper look behind the short version — what NNMT does and why blocking it was interesting, what the mouse studies measured and where they stopped, the oral-capsule question, and what your consultation covers.

What it is

5-Amino-1MQ is not a peptide, despite being sold and marketed alongside them. It is a small synthetic chemical designed in a University of Texas laboratory in 2017 to block an enzyme called nicotinamide N-methyltransferase (NNMT), an enzyme that is unusually active in the fat tissue of obese animals.

It is sold as an oral capsule for fat loss and metabolic support, positioned as a once-daily supplement rather than an injectable therapy.

What people use it for

USED FOR	EVIDENCE	WHAT THAT MEANS
Fat loss and weight loss	**Animal data only**	The original finding: obese mice given it for eleven days lost weight without eating less.
Insulin sensitivity and metabolic health	**Animal data only**	Same line of mouse work — the enzyme it blocks is unusually active in obese fat tissue.
Cellular energy and NAD+	**Animal data only**	Blocking the enzyme leaves more of the NAD+ building block available, shown in animals.

How it works

NNMT is an enzyme that uses a molecule called SAM (S-adenosylmethionine) to inactivate nicotinamide, a building block the body otherwise uses to make NAD+, the coenzyme cells rely on for energy production. NNMT activity rises in the fat tissue of obese animals, and researchers proposed that blocking the enzyme might redirect nicotinamide back toward NAD+ production and away from fat storage.

5-Amino-1MQ was engineered specifically to inhibit NNMT. In the mouse studies that established the compound's reputation, blocking NNMT was associated with weight loss without a reduction in food intake, along with markers suggestive of improved insulin sensitivity — findings researchers attributed to downstream effects on NAD+ availability and fat-cell metabolism.

What the research actually looked at

The finding most often cited for 5-Amino-1MQ comes from a study in obese mice: animals given the compound for eleven days lost weight without any change in how much they ate. That single line of mouse research is the foundation for essentially every claim made about the compound.

The same research group's follow-on work looked at insulin sensitivity and NAD+-related markers in animal models, using the same NNMT-blocking mechanism. No human clinical trial

of 5-Amino-1MQ has been published — for fat loss, metabolic health, or any other proposed use. Everything currently claimed about the compound in people is extrapolated from that animal data rather than demonstrated in it.

Common misunderstandings

- **“It's a peptide.”** It is a small synthetic molecule, not a peptide, despite being sold and marketed alongside peptide products.
- **“The mouse data means it works the same way in people.”** No human trial of 5-Amino-1MQ for fat loss or metabolic health has been published; the entire evidence base is animal research.
- **“Because it's oral, it must be gentler or safer than an injectable.”** The animal studies that generated interest in this compound used injection; the oral capsule form sold today has not been tested in the same studies, and how well it is absorbed by mouth is unclear.
- **“It's a proven NAD+ booster.”** The proposed NAD+ mechanism is inferred from animal biology; no human study has measured whether it raises NAD+ levels in people.

Who it is not for

- Anyone who is pregnant, might become pregnant, or is breastfeeding.
- Anyone with a current or previous cancer, without their oncologist's input.
- Anyone with liver disease or abnormal liver tests.
- Anyone taking prescription medication, without a review first — there is no human interaction data.

Competitive athletes: 5-Amino-1MQ is not currently named on the WADA Prohibited List. Athletes in tested sport should still confirm current status with their sport's anti-doping authority before use, since compounded and supplement-market products carry a contamination risk that can result in an inadvertent positive test regardless of the labeled ingredient.

Who makes this, and under what rules

Everything dispensed here is prepared by a licensed 503A compounding pharmacy — not bought from a supplier and repackaged, and not sourced from an online seller.

That designation carries specific, enforceable obligations:

- **A prescription in your name.** Compounded from a prescription for one identified patient, after an evaluation.
- **Licensed and inspected.** A state licence, subject to inspection by the state board of pharmacy.
- **Ingredients that can be traced.** A certificate of analysis from an FDA-registered facility, back to a named source and lot.
- **Sterile preparation to a published standard.** USP standards.
- **A paper trail.** A lot number and beyond-use date on every preparation.

A compounded preparation is not an FDA-approved medicine — but it is the difference between a licensed pharmacy working to a standard and an unregulated website.

Where this stands

Never approved, and not actually a peptide. A small synthetic molecule despite being sold alongside peptides.

The evidence is animal and cell work. The interest comes from mouse studies of an enzyme involved in fat metabolism. No published human trial has measured whether it changes body composition in people.

Everything claimed rests on rodent research.

Clinical detail

Designed in 2017 in a University of Texas laboratory as a small-molecule NNMT (nicotinamide N-methyltransferase) inhibitor.

Not a peptide — a synthetic small molecule, despite common marketing alongside peptide products.

Foundational data: obese mice given the compound for eleven days lost weight without reduced food intake.

Sold as an oral capsule; the animal research establishing its effects used injection.

No published human clinical trial exists for any proposed use.

The information on this page is general education. It is not medical advice and it is not a recommendation for any treatment.

5-Amino-1MQ is not an FDA-approved medicine for the uses described here. A compounded preparation is not an FDA-approved product and has not been reviewed by the FDA for safety, effectiveness or quality.

Descriptions of what this is used for reflect how it is commonly used and discussed. They are not claims that it is effective for any purpose.

Any prescription follows an individual evaluation by a licensed provider and is issued only where that provider judges it clinically appropriate. A prescription is never guaranteed.